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AQA A Level Further Mathematics (7367)

Topic DC: Network Flows

Sub topic DC1: Interpret Directed Networks

Mark Scheme

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Mark Scheme

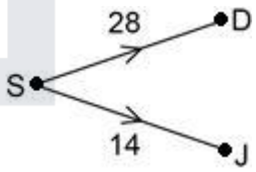
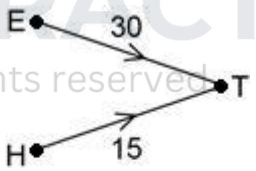
Topic	**DC: Network Flows**
Sub topic	**DC1: Interpret Directed Networks**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

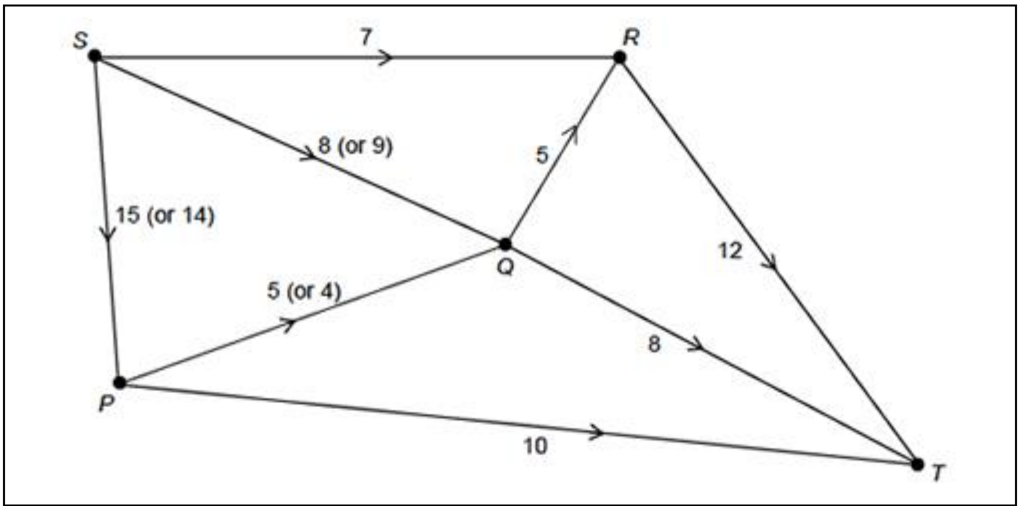
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Finds the value of the cut	AO1.1b	B1	37
(ii)	Deduces a weak inequality for the value of max flow based on their value of the cut Must see ≤ their value of the cut OE	AO2.2a	B1F	Max flow ≤ 37
(b) (i)	Lists both sources CAO	AO1.1b	B1	D and J
(ii)	Adds supersource S with directed arcs to D and J labelled with weights at least 28 and at least 14 respectively	AO1.1b	B1	
(c) (i)	Lists both sinks CAO	AO1.1b	B1	E and H
(ii)	Adds supersink T with directed arcs to E and H labelled with weights at least 30 and at least 15 respectively	AO1.1b	B1	
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Determines correctly the value of both cuts	AO1.1b	B1	35 and 36
(b)	Deduces the maximum flow in ‘their’ network by use of the maximum flow-minimum cut theorem and explains their reasoning.	AO2.4	E1	Max flow = 30 as max. flow = min. cut and the value of the minimum cut is 30



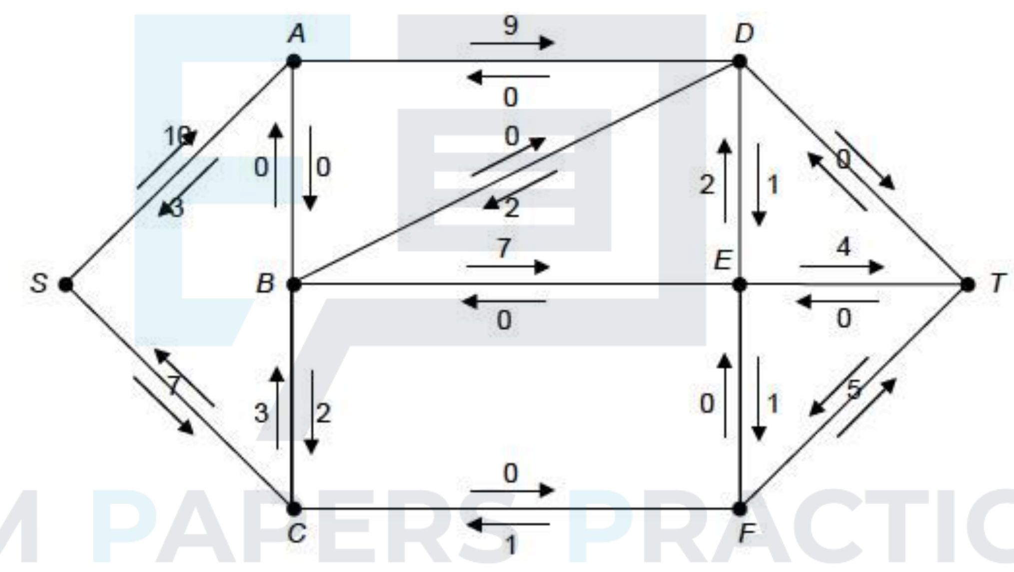
(c)	Determines correctly the flow along the saturated arcs	AO1.1b	B1	*SR* = 7, *QR* = 5, *RT* = 12, *QT* = 8, *PT* = 10
	Determines correctly a possible flow for each of *SP*, *SQ* and *PQ*	AO1.1b	B1	*SP* = 15 and *SQ* = 8 and *PQ* = 5 or *SP* = 14 and *SQ* = 9 and *PQ* = 4
				Total 4 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Explains correctly using maximum flow into and minimum flow out of *A*	AO2.4	E1	Max flow into vertex *A* = 7 Min flow out of vertex *A* = 4 + 3 = 7 As flow into *A* = flow out of *A*, both *AD* and *AB* must be at the lower capacity
(ii)	Explains correctly using maximum flow into and minimum flow out of *E*	AO2.4	E1	*AD* = 4 from (a)(i), so *DE* = 1 and *DT* = 3. Since flow out of vertex *E* is at least 2, and *DE* = 1, *BE* must be at its upper capacity of 1.
(b)	Explains correctly the statement using a cut in the network	AO2.4	E1	Min flow across the cut {*S*, *A*, *B*, *C*}/{*D*, *E*, *G*, *T*} = *AD* + *BE* + min(*BG*) + min(*CG*) = 4 + 1 + 5 + 2

			= 12 > 11

(c) (i)	Finds a potential increase and decrease for each arc with a value on the two arrows for each arc, with *SA*, *SC*, *AB*, *BC* and *CF* correct.	AO1.1a	M1	See diagram below
	Determines correctly the potential increase and decrease for each arc with values on all arrows correct.	AO1.1b	A1	



(ii)	Finds correctly one augmenting path and flow	AO1.1b	B1	**Augmenting Path** / **Flow** *SADEFT* / 1 *SCBET* / 3 *SADBET* / 1
	Finds correctly two augmenting paths and the flows	AO1.1b	B1	
	Finds correctly three (or four) augmenting paths and the flows, and clearly states the maximum flow in context with units	AO3.2a	B1	Maximum flow through the network is 22 litres per second
(iii)	Constructs a rigorous mathematical proof using the value of a cut and the maximum flow-minimum cut theorem (to achieve this mark, the student must clearly show the cut they are calculating the value of **and** clearly state that the value of this cut is equal to the value of the	AO2.1	R1	{*S*, *A*, *B*, *C*, *D*, *E*}/{*F*, *T*} = 7 + 3 + 9 + 3 = 22 As the flow (22) is equal to the value of the cut (22), the maximum flow is 22 by the maximum flow-minimum cut theorem.

flow in **(c)(ii)** and then conclude that the flow is maximal by the maximum flow-minimum cut theorem)			

(iv)	Argues that, as 7 litres per second are initially flowing through node *E*, only a further 2 litres per second can flow through node *E*	AO2.4	R1	Flow through network can only increase by 2 litres per second as *DT* and *CF* are already saturated Therefore restricted capacity node reduces the maximum flow to 17 + 2 = 19 litres per second
	Interprets the impact of the restricted capacity node, concluding that the maximum flow is reduced to 19 litres per second	AO3.A	R1	
				Total 11 marks

Q4.

(a) (i) Max Flow = 50
(Min cut = 50)
Either statement

E1

(ii) 35 ≤ max flow ≤ 50
(or min cut)
E1 for strict inequalities

E1, E1

(iii) Error or contradiction
oe

E1

4

(b) At F,

flow in ≥ 8
flow out ≤ 7

Stating F and one of the 'flows'

M1

A1

2

[6]

Q5.

(a) $ABEH$ 8

$ACFH$ 5

$ADGH$ 11

B1

1

(b) (i) $ACEH$ 2

One correct route and flow

M1

$ACGH$ 4

At least one other correct

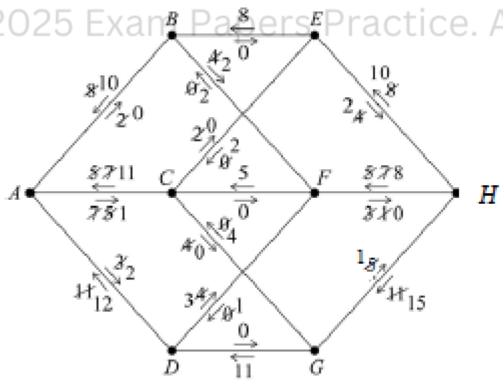
A1

Either $ADFH$ 1 and $ABFH$ 2

All correct

A1

Or $ADFH$ 3



Forward and back flows on diagram

M1

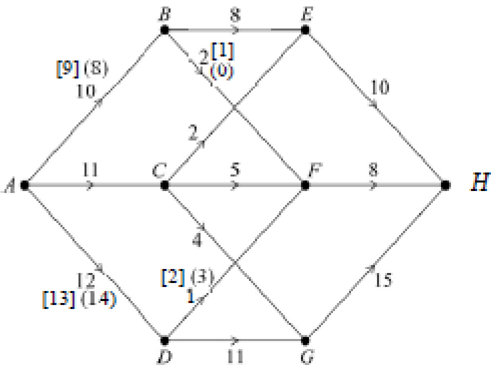
All correct

A1

5

(ii) Max flow 33

B1



OE

B1 2

(c) Cut through BE, CE, FH, CG, DG

B1 1

[9]

Q6.

(a) 19

B1 1

(b) E

B1 1

(c) C

B1 1

(d) $x = 8$
$y = 13$
$z = 39$

B1 x 3 3

(e) 76

B1 1

(f) 83

B1 1

[8]

Q7.

(a) (i) 18 +(0+) 10 + 3 + 5 (= 36)

B1

1

(ii) 30, 32, 36 (missing cut values)

B1 each value correct

B3

3

(iii) Max flow = 29

Award B0 E1 if their min^m cut is < 29

B1

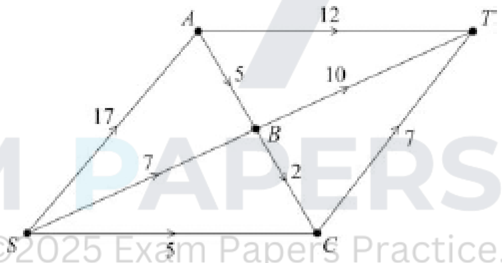
because value of minimum cut is 29

and min^m value explained as max flow

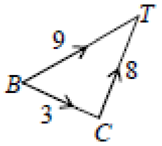
E1

2

(iv)



may have



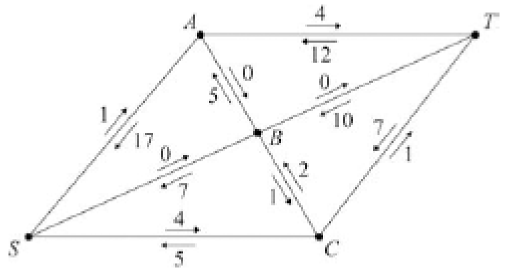
see alternative solution on next page

B1

cao

2

(b) (i)



potential flow (forward and back)
4 pairs 'correct' including SC and AT
ft their (a)(iv) provided 0 < flow < 30

M1

all pairs correct (condone missing 0s)
ft their (a)(iv) if correct flow < 29

A1 ✓

one correct flow in table

m1

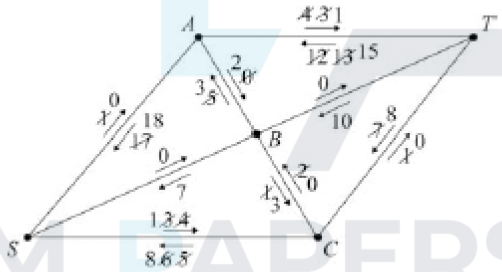


table correct	*SAT*	1
	SCT	1
	SCBAT	2

If (a)(iv) flow < 29 then may score A1 for correct table giving max flow of 33

A1

(see also the alternative solution)
modifying flows (forward and back)
1 flow correct ft their initial flow

m1

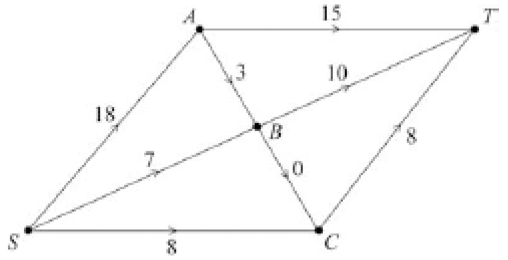
modified flows all correct, including all 0s (may score A1 from a correct flow < 29 seen in (a)(iv) if final flow correct)

A1

6

(ii) new max flow = 33

B1



6 flows correctly interpreted from their labelling procedure provided M2 or M3 scored in (b)(i)
(may have AB 2, AT 16, BT 9 − see over)

M1

flow correct
***SC** B1 if flow of 33 shown correctly but not from correct labelling procedure*

A1

3

[16]

Q8.

(a) Arrival gates are U and R

B1

1

(b) Cut value = 45 + 53 + 20 + 37 + 0
= 155

B1

1

(c) Max flow along $UTSP$ is 17

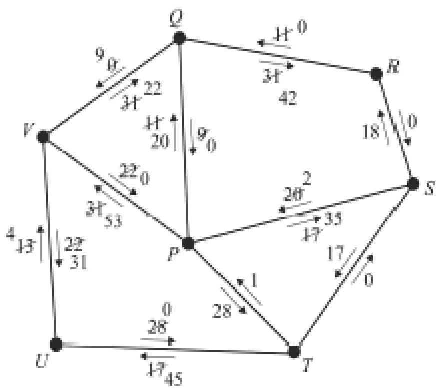
B1

and along $RQVP$ is 31

B1

2

(d) (i)



Route	Value of Flow
UTSP	17
RQVP	31
RSP	18
RQP	11
UTP	28
UVP	22
UVQP	9

Initial flows along *UTSP* and *RQVP* with potential increases and decreases

B1

Table: first route and correct flow

After UTSP and RQVP

M1

Another route and flow

A1

Table correct

A1

Network: attempt to use labelling procedure with forward/backward flows

M1

All diagram correct

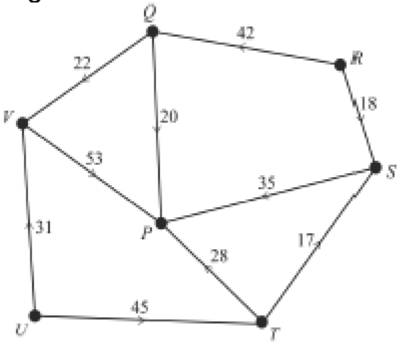
A1

6

(ii) Maximum flow = 136

B1

Figure 2 correct:



Other possible answers

B1

2

(e) Rate reduced by 3

"their" maximum flow – 3

M1

New maximum is 133

A1

2

[14]

Q9.

(a) D

B1

1

(b) (17 + 25 + 35 + 13 + 12 + 13 = 115)

B1

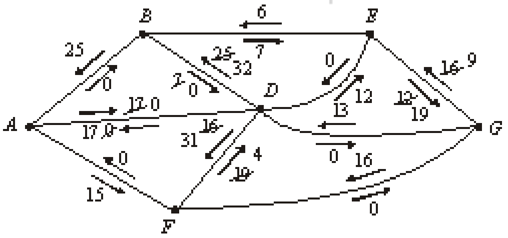
1

(c) ABD_{max} = 25 ; GED_{max} = 12

B1B1

2

(d) (i)



Forward and backward flows

M1

Adjusting flows on diagram

M1

Routes and flows in chart

M1

One correct other than ABD, GED

A1

Another correct

A1

Route	ABD	GED	GFD	GD	AD	AFD	GEBD
Flow	25	12	16	13	17	15	7

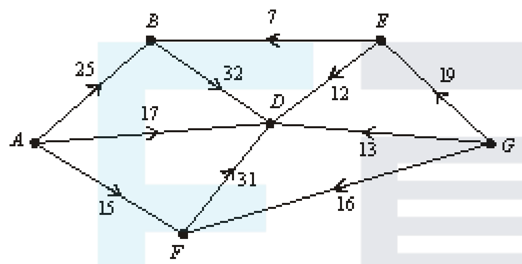
All correct

A1

6

(ii) Total = 105
Max flow

B1



B1

2

(iii) Cut through AF, AD, BD, DE, DG, and GF

Through 3 saturated arcs (fairly generous)

M1

Correct

A1

2

(e) Reduce max flow by their EG
changing 19 to 15

Reduce by 4 since everywhere else saturated

M1

$\Rightarrow$ New max = 101

Correct answer $\Rightarrow$ *2 marks*

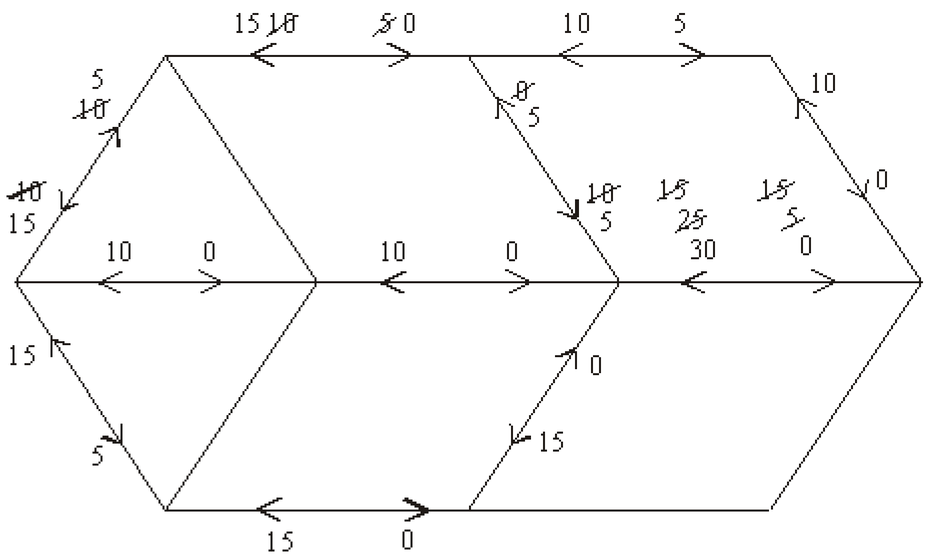
A1

2

[16]

Q10.

(a)



SCA

first flow M1

second flow A1

A1

∴ maximum flow = 40

B1

4

(b) Cut BE, DG, CF

B1

1

(c) Out of A – max 50

B1

∴ from A must be road

Same as J

B1

∴ join AJ direct

2

[7]

Q11.

(a) A, B, C, D

B1

1

(b) *I*

B1

1

(c) 20 + 20 + 10 + 10 + 40 + 50 + 30

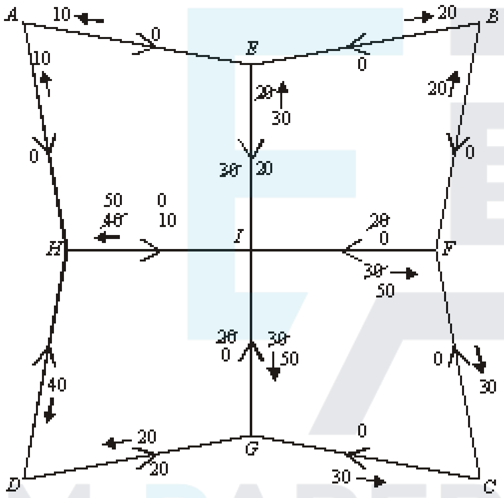
M1

= 180

A1

2

(d)



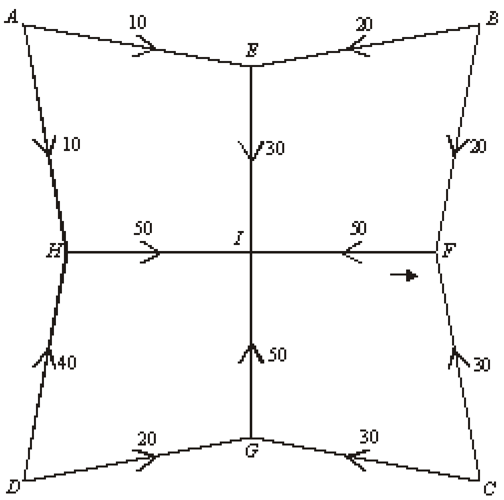
SCA

M1

1 complete flow

M1

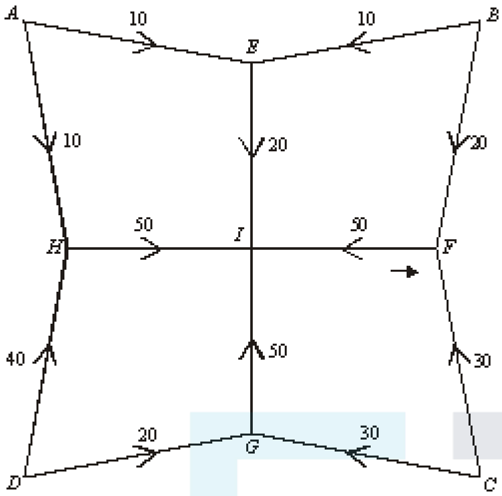
Final:



OE

A1

(e)



Their max – 10

M1

Max flow = 170

CAO

A1

2

[11]

Q12.

(a) (i) Source *A* , *C*

B1

1

(ii) Sink *G*, *I*

B1

1

(b) (i) Flow = 100

B1

1

(ii) $x = 14$

B1

1

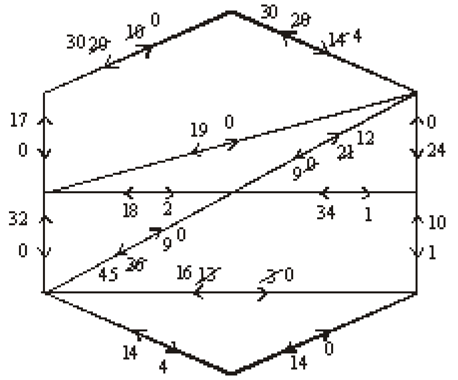
(iii) $y = 34$

B1

1

(c) Flow = <u>122</u>

B1



Initial diagram

SCA M1

1 flow; OE M1

A1

2nd flow M1

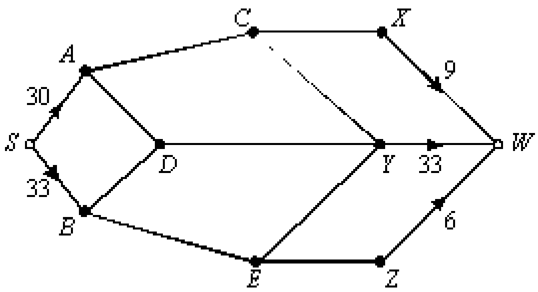
All correct; OE A1

6

[11]

Q13.

(a)



2

for S and W M1

for five arcs correct (≥) A1

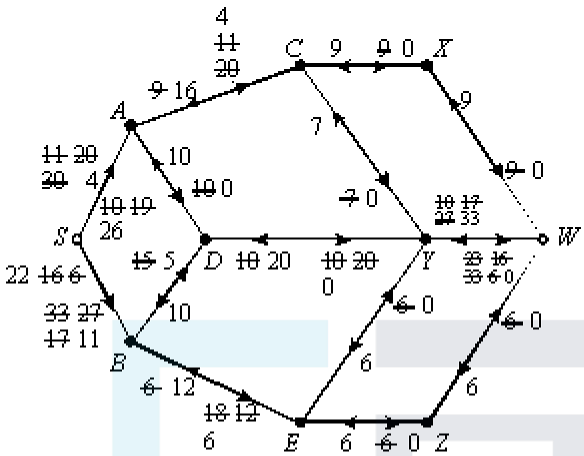
(b) *SADYW* = 10 B1

SBEZW = 6

B1

2

(c) (i)



for starting with SADYW, SBEZW

M1

SCA

M1

for AC

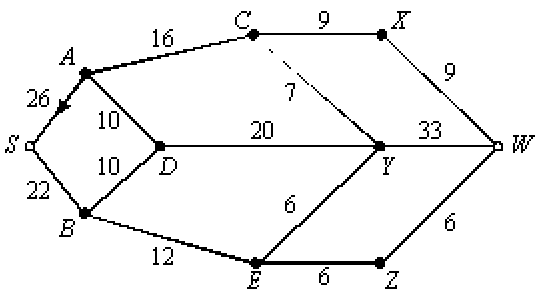
A1

for SA

A1

for SB

A1



OE

B1

6

(ii) Maximum flow = 48

M1

Minimum cut *CX*, *YW*, *EZ*
(9) (33) (6)

= 48

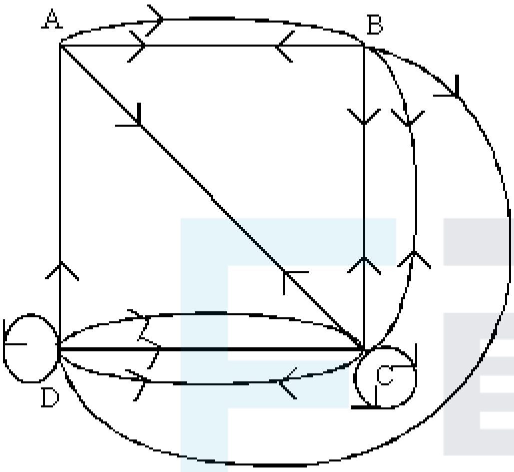
A1

2

[12]

Q14.

(a)



Directed graph with one correct

B1

C→C, D→ correct

B1

Fully correct

B1

3

(b) Not symmetrical

OE

E1

1

[4]

Q15.

(a) ***SC, SB, AT***

M1 A1

2

(b) (i) SCT 1

M1 A1

2

(ii) $SABCT$ 2

M1 A1 **2**

(c) any flow ≤ any cut
so any flow ≤ 16

B1 **1**

(d) Choose AT.

B1

The arc must be in the minimum cut in

M1

(a). Also it must be in $\{AT, BT, CT\}$
which is a cut of 17.

A1 **3**

[10]

Q16.

(a) e.g. $SABT$ 4
$SACBT$ 2

A1

$SCDT$ 2

A1

$SEDFT$ 3

A1

$SEFT$ 3 (total 14)
Any two paths
Remaining paths to make total 14
(or 2 for cao)

A1 **5**

(b) EF, ED, CD, BT
(3 + 3 + 2 + 6 = 14)

M1 A1 **2**

(c) (i) Flow in SE ≤ flow out at E ≤ 3 + 3

B1

1

(ii) Flow into C = flow out of $C \leq 3 + 2 = 5$

M1

So if AC has a flow of 4 then SC has a flow of at most 1.

A1

2

(iii) If AC is saturated the maximum flows in SA, SC and SE are 6 (its capacity), 1 (by part (ii)) and 6 (by part (i)). So if AC is saturated the maximum flow out of S is 6 + 6 + 1 = 13 which is less than the unrestricted maximum flow.

M1

A1

2

[12]

Q1.

The majority of students determined the value of the cut correctly, with the most common errors being numerical slips. Only 51% of students correctly interpreted their value of the cut, with many stating generic comments about the maximum-flow minimum-cut theorem but not making the connection between the value of any cut and the implications for the maximum flow.

Most students spotted the sources and sinks of the network, the most common error being to include a third node which was not a source or sink. However, less than half of the students correctly added arcs to the diagram, with common mistakes being the lack of arrows or correct capacities on each arc.

Q4.

This question proved to be a challenge for students. Students were required to think of the implications of statements and not merely apply an algorithm. Part (a)(i) was answered quite well, but in part (a)(ii), many students failed to state a range for the maximum flow. Some students realised that there had to be an error in part (a)(iii). Fully correct answers to part (b) were rarely seen. Many students highlighted the problem at a vertex other than 'F' and justified their answer by considering maximum flow in and minimum flow out.

Q5.

Most students answered part (a) correctly. Although there were many correct solutions for part (b)(i), a significant number of students failed to give a convincing solution on their diagram. It is essential that students use ‘backward’ arrows to show existing flows and ‘forward’ arrows to show potential flows. There was a significant number of students who scored 5 marks in part (b)(i) but failed to score both marks in part (b)(ii). Part (c) was answered poorly. Students were asked to list the edges of their cut; many simply drew a cut on their diagram.

Q6.

Although many students scored the majority of marks on this question, full marks were rarely seen. Errors were common in answers to parts (e) and (f).

Q7.

(a) (i) Many students were able to show the correct working to give the value of the cut.

(ii) Far fewer were successful in finding the three values of the remaining cuts in the table.

(iii) Some thought that the “maximum” cut was needed to give the maximum flow and others wrote “maximum flow ≤ minimum cut”. The 29 value had been given to help students and this proved valuable for those who had been unable to find the 3 correct values in the table.

(iv) Those with a maximum flow of 29 were usually able to draw correct flows along the given edges, although some gave at least one value greater than the capacities on the original network.

(b) Some missed the vital point of supplementing the flows by 4 along *SC* and *AT* and made no further progress. It was good to see many students trying to set out their solution in a logical manner and once again the diagram and table clearly helped. There are still many students unfamiliar with the "labelling procedure" who fail to show both potential forward **and** backward flows on their network. Students are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit students for their original values if they have been obliterated during augmentation.

Some credit was given to students who had used flow augmentation correctly but had failed to find a maximum flow equal to 33, if they interpreted their flows correctly on the final diagram.

Q8.

In part (a) most candidates identified the arrival gates and scored an easy mark.

In part (b) quite a few candidates made errors in calculating the value of the cut, with common wrong answers being 138 and 172.

In part (c) the values of the initial maximum flows were usually correct.

In part (d) it was good to see candidates trying to set out their solution in a logical manner and once again the insert clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made.

For part (d)(ii) the incorrect answer of 127 was seen far more often than the correct value for the maximum flow, since candidates seemed unaware of the need to introduce an additional flow of 9 by actually reducing the flow along some edges that were previously at full capacity.

In part (e) those who had the correct maximum flow in part (d) were usually successful in finding a new maximum flow and those who did not have a correct answer were given some credit for reducing their previous flow by 3 or explaining how the extra 3 passengers could be accommodated in some way.

Q9.

Part (a) was usually correctly answered but a number made errors in calculating the value of the cut in part (b). The values of the initial maximum flows in part (c) were usually correct also. The insert **Figure 4** was intended to help candidates to set out their solution to part (d) in a logical manner. Some candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. It should be clear to the examiner what the previous values were when such modification is made and the final backward flows should be the values transferred onto **Figure 5** to give a possible maximum flow. Very few were successful in finding a cut of the same value as the maximum flow. The final part (e) was well answered by most of those who had found the correct maximum flow.

Q10.

Good candidates were able to score the majority of marks on this question, with clearly labelled diagrams. Weaker candidates failed to score the marks in part (a). Candidates should be encouraged to list the paths that they are augmenting at each stage. Part (c) was the least well answered question on the paper.

Q11.

This was the first time a question had been set with multiple sources. Parts (a) and (b) were, in general, well answered. Nearly all candidates were able to find the value of the cut X in part (c). In part (d)(i) candidates only scored full marks if they showed each separate flow augmentation.

Q12.

More able candidates were able to score maximum marks on this question, with clearly labelled diagrams. Weaker candidates failed to score the marks in parts (b) and (c). There was a lack of understanding of 'an initial flow' and in flow augmentation candidates concentrated their efforts on either the sources or the sinks but not both. Candidates should have listed the paths that they were augmenting at each stage. This would have helped them to establish the additional flow of 22 units that was required in this question.

Q13.

Candidates were required to add a super source and a super sink to the given diagram. This was well answered although a number of candidates did not include values on the arcs that at least equalled the total values at points A and B. Part (b) was well answered and to assist candidates they were expected to take these as initial flows in part (c) before using flow augmentation to the diagram. There were many excellent solutions to this question, with working shown on the diagrams, together with a clearly labelled final solution.

Q14.

This question was a straightforward graph theory question. Unfortunately, a number of candidates thought that the numbers represented lengths and they then stated that the graph must be directed if the distance from A to B was different to the distance from B to A.